

Skill-Guided Reconstruction Audit: Fukami et al. Super-Resolution of Turbulent Flows

With-skill benchmark arm: gpt-5.5, xhigh reasoning, Codex exec

July 1, 2026

Abstract

This manuscript seed reconstructs the argument of Fukami, Fukagata, and Taira’s 2019 super-resolution study using the benchmark dense summary plus the CFD-AI package guidance for scientific journal writing, claim-evidence auditing, CFD reproducibility, and vocabulary control. The defensible claim is narrow: convolutional and hybrid downsampled skip-connection/multi-scale models reconstruct high-resolution velocity or vorticity fields from low-resolution inputs for a cylinder wake at $Re_D = 100$ and two-dimensional homogeneous decaying turbulence, with bicubic interpolation as a classical baseline. The source scope does not support exact error values, layer counts, solver settings, train/test seed details, full hyperparameter tables, final figure captions, code repository status, real-time deployment, solver acceleration, or universal turbulence recovery; these remain TODO rather than implied.

1 Source Scope and Claim Boundary

The reconstruction uses three evidence sources: the benchmark task prompt, the dense source summary, and the local gold-paper note for Fukami et al. The gold note reports the paper as Super-resolution reconstruction of turbulent flows with machine learning, *Journal of Fluid Mechanics* 870, 106–120, DOI 10.1017/jfm.2019.238; it also states that full PDF figure-caption verification remains TODO.

The one-sentence claim is:

Fukami et al. formulate low-resolution to high-resolution flow-field reconstruction as a supervised learning problem and test convolutional and hybrid multi-scale models on a cylinder wake at $Re_D = 100$ and two-dimensional homogeneous decaying turbulence, comparing against bicubic interpolation and flow-field/statistical diagnostics under the supplied source scope.

This is a reconstruction claim, not a turbulence-closure, deployment, or solver-acceleration claim.

2 Claim-Evidence Map

3 Method Reconstruction

Let $\mathbf{q}_\ell$ be a low-resolution field produced by a documented downsampling operation and $\mathbf{q}_h$ be the corresponding high-resolution reference field. The supervised reconstruction problem is

$$\hat{\mathbf{q}}_h = \mathcal{N}_\theta(\mathbf{q}_\ell), \quad \mathcal{L}(\theta) = \|\hat{\mathbf{q}}_h - \mathbf{q}_h\|_2^2. \quad (1)$$

Claim	Status		Evidence location	Required TODO
Low-resolution flow fields are mapped to high-resolution fields	Supported scope	with	Dense summary and gold note describe the reconstruction task	Exact normalization and input and output tensor construction
CNN and hybrid DSC/MS models are used	Supported scope	with	Dense summary identifies both model families	Layer counts, filter widths, parameter counts, architecture diagram verification
Cylinder wake at $Re_D = 100$ is a demonstration case	Supported		Dense summary and gold note	Solver, mesh, time step, boundary/initial conditions
Two-dimensional homogeneous decaying turbulence is a stress test	Supported scope	with	Dense summary and gold note	Initial condition, resolution, statistical sampling, exact diagnostic panels
Bicubic interpolation is a baseline	Supported		Dense summary and gold note	Same-data metric table and exact numerical comparison
Small-data behavior includes as few as 50 snapshots	Partial		Dense summary says accessible text notes this claim	Exact curve values, split protocol, variability across runs
High-wavenumber or statistical recovery is demonstrated	Partial		Dense summary says physical/statistical diagnostics matter, exact panels TODO	Verified spectra/statistics and caption-level evidence
The method is robust, generalizable, or real-time	Unsupported as a broad claim	a	Not supplied	Noise, out-of-distribution, deployment, and hardware evidence before using these terms

Table 1: Claim-evidence map following the package claim-audit rubric. Unsupported claims are converted to TODOs.

The supplied sources support CNN and hybrid downsampled skip-connection/multi-scale model families, average and max pooling in the downsampling protocol, Adam optimization, and early stopping. The sources do not supply enough evidence to reconstruct all architecture, optimizer, normalization, or split details.

4 Evidence Sequence and Figure Roles

The figure sequence should prove progressively stronger claims while preserving TODO boundaries.

5 Reviewer-Risk Audit

6 Implications for a New Paper in This Line

A follow-on CFD-AI paper should not merely add a larger neural network. The defensible experiment matrix should include:

Reproducibility item	Present?	Evidence	Blocking TODO
Governing equations	Partial	Gold note says incompressible Navier–Stokes for the wake and 2D vorticity transport for turbulence are in accessible text	Exact equation statement and assumptions from paper
Regime numbers	Partial	$Re_D = 100$ for cylinder wake is supplied	Other nondimensional groups or turbulence parameters
Geometry/domain	Partial	Circular-cylinder wake is identified; homogeneous turbulence case is identified	Domain dimensions and coordinate conventions
Boundary/initial conditions	Missing	Not in dense summary	Wake BC/IC and turbulence initial condition/forcing
Grid and time integration	Missing	DNS is stated, but implementation details are not supplied	Mesh, time step/CFL, schemes, convergence checks
Data splits	Partial	Train/validation/test separation is indicated in gold note	Exact split definition, seed, leakage prevention
ML training	Partial	L2 objective, Adam, early stopping are noted	Learning rate, batch size, epochs, normalization, seeds, runs
Baselines	Partial	Bicubic interpolation is named	Full baseline matrix and same-metric tables
Compute and code/data	Missing or unknown	Code availability statement appears in accessible arXiv text, but repository status is unknown	Verified URL, commit/version, commands, hardware/runtime

Table 2: CFD reproducibility ledger. The manuscript should not fill missing numerical-method details from expectation.

- same-data comparisons against interpolation and simple neural baselines;
- explicit spatial downsampling factors and train/validation/test splits;
- reconstruction metrics plus physical diagnostics such as vorticity PDFs, spectra, conservation residuals, or force histories where relevant;
- stress tests over snapshot count, coarse resolution, noise, and sparse observations;
- holdout tests on a named axis before claiming generalization;
- hardware, runtime, memory, and code/data instructions before claiming efficiency or deployability.

Pipeline and downsampling. Evidence role: defines the low-resolution input, high-resolution target, and reconstruction operator. TODO: exact pooling factors, tensor dimensions, and architecture schematic.

Figure 1: Reconstruction pipeline placeholder. The caption states the evidence role rather than implying unavailable implementation details.

Cylinder wake at $Re_D = 100$. Evidence role: compares ground truth, coarse input, bicubic interpolation, and learned reconstruction for velocity or vorticity fields. TODO: exact color scales, error values, and panel order.

Figure 2: Cylinder-wake reconstruction placeholder with explicit claim boundary.

7 Conclusion

The package-guided reconstruction remains narrower than a fluent generic paper. It identifies the paper’s CFD/SciML role, preserves the supported low-resolution to high-resolution contribution, and makes missing solver, architecture, metric, and figure-caption details visible. This is the desired failure-avoidance behavior for scientific paper reconstruction: unknown facts stay unknown, while the manuscript still retains the physical problem, ML transformation, evidence sequence, and reviewer-defense structure.

References

- [1] K. Fukami, K. Fukagata, and K. Taira. Super-resolution reconstruction of turbulent flows with machine learning. *Journal of Fluid Mechanics*, 870:106–120, 2019. DOI: 10.1017/jfm.2019.238.

Training-snapshot and PDF diagnostics. Evidence role: tests data amount and checks distributional agreement for vorticity in the cylinder wake. TODO: exact snapshot-count curve, uncertainty, and PDF values.

Figure 3: Data-sensitivity and vorticity-PDF placeholder.

Two-dimensional homogeneous decaying turbulence. Evidence role: stress-tests reconstruction of smaller-scale turbulent structure from coarse inputs. TODO: verified spectral/statistical panels and quantitative values.

Figure 4: Turbulence stress-test placeholder.

Risky wording or gap	Why it is unsafe	Safer manuscript action
“Robust reconstruction”	Robustness needs an identified stress axis such as noise, sparsity, OOD regime, or coarse-grid severity	Say “reconstruction under the tested downsampling conditions” until stress tests are supplied
“Generalizable turbulence recovery”	The supplied cases do not prove unseen geometry, 3D turbulence, or experimental transfer	Limit to cylinder wake and 2D homogeneous decaying turbulence
“Physically consistent”	Requires a named physical diagnostic, not only visual similarity	Name vorticity PDF and any verified spectrum/statistical diagnostic; keep exact panels TODO
“Real-time” or “efficient”	Requires hardware, batch size, runtime, and a comparator	Mark runtime and deployment claims TODO
Exact numerical superiority	Exact values are absent from the source summary	State comparison structure, not unverified margins

Table 3: Vocabulary and claim-calibration audit.